

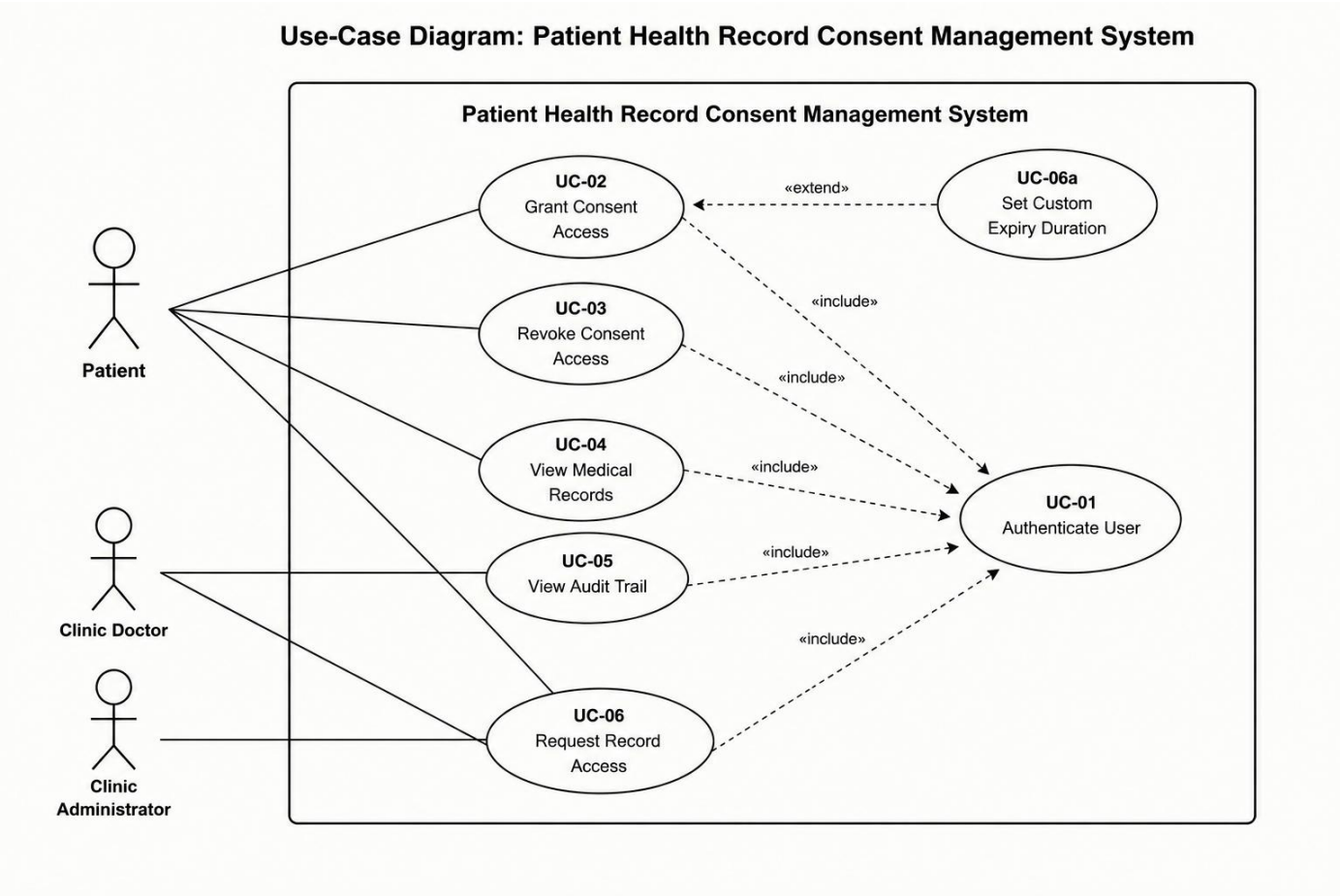
Patient Health Record Consent Management System

Software Engineering Lab 1 — Requirements Engineering & UML Use-Case Modelling

Requirements Table

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall allow patients to grant time-bounded (e.g., 24-hour) access permissions for specific diagnostic records to verified clinic doctors.	High	Pass: Clinic access automatically revokes upon expiry timestamp. Fail: Clinic accesses records after consent expiration.	Core function enabling patient-controlled, time-limited data sharing.
FR-002	Functional	The system shall allow a patient to revoke a previously granted access permission at any time before its scheduled expiry.	High	Pass: A clinic access attempt is denied within 5 seconds of the patient tapping "Revoke". Fail: Clinic can still view the record after revocation.	Patients must retain immediate, continuous control over who can view their health data.
FR-003	Functional	The system shall require a clinic doctor to submit an access request specifying patient ID, record type, and reason for access before consent can be granted.	Medium	Pass: A request submitted without a reason field is rejected with an error message. Fail: A request lacking a reason is accepted and forwarded to the patient.	Capturing purpose-of-access supports informed patient decisions and audit accountability.
FR-004	Functional	The system shall send the patient a notification (in-app, SMS, or email) at least one hour before an active consent grant expires.	Medium	Pass: Notification logs show a reminder sent 55–65 minutes before every grant's expiry timestamp. Fail: No reminder is logged, or it falls outside that window.	Gives patients a timely opportunity to extend or let access lapse before it expires.
FR-005	Functional	The system shall allow a patient to view a complete, filterable audit trail of all consent grants, revocations, and record accesses linked to their account.	High	Pass: Filtering by a date range returns all matching events within 3 seconds. Fail: Events are missing or the query fails.	Transparency into access history builds patient trust and supports regulatory compliance.
NFR-001	Nonfunctional (Performance & Security)	All consent grant, revoke, and access events shall be permanently written to an append-only audit trail.	High	Pass: Benchmarking tests confirm target latency and security standards under simulated peak load.	Guarantees a tamper-evident record of every consent-related action for compliance and dispute resolution.
NFR-002	Nonfunctional (Security)	All patient health records and consent data shall be encrypted using AES-256 at rest and TLS 1.2 or higher in transit.	High	Pass: Security scan confirms no record is stored or transmitted unencrypted. Fail: Any record is found in plaintext.	Protects sensitive health data from unauthorized disclosure and supports healthcare data-protection requirements.

UML Use-Case Diagram



Use-Case Flow Specification

Patient Health Record Consent Management System

Use Case ID	UC-06	Use Case Name	Request Record Access
Primary Actor	Clinic Doctor	Supporting Actors	Patient; System (Notification Service, Audit Trail Service)
Includes	UC-01: Authenticate User	Purpose	Submit a structured request for patient record access before consent is granted.

Preconditions

- The clinic doctor has a registered and verified account.
- The patient exists in the system and can be identified using the patient ID.

- The required medical/diagnostic record type is available in the system.

Postconditions

- A valid access request containing patient ID, record type, and reason is submitted.
- The request is sent to the patient for consent consideration.
- The access-request event is recorded in the audit trail.

Main Success Scenario

1. Clinic Doctor logs in and selects “Request Record Access” from the dashboard (includes UC-01: Authenticate User).
2. System displays a form requesting Patient ID, record type, and reason for access.
3. Doctor enters the Patient ID.
4. System validates that the patient can be identified.
5. Doctor selects the required record type.
6. Doctor enters a clear reason for requesting access.
7. Doctor submits the access request.
8. System validates that all required fields are complete and valid.
9. System records the request in the audit trail.
10. System sends the access request to the patient for review and consent.
11. System displays a confirmation message to the doctor.
12. Use case ends successfully.

Alternate Flow — 8a: Reason Not Provided

8a1. System detects that the reason field is empty. **8a2.**

System displays “Reason for access is required.”

8a3. Doctor enters a valid reason.

8a4. System validates the request and continues from Step 9.

Requirement Traceability

This use case implements FR-003: the clinic doctor must submit an access request containing patient ID, record type, and reason before consent can be granted. A missing reason causes the request to be rejected.